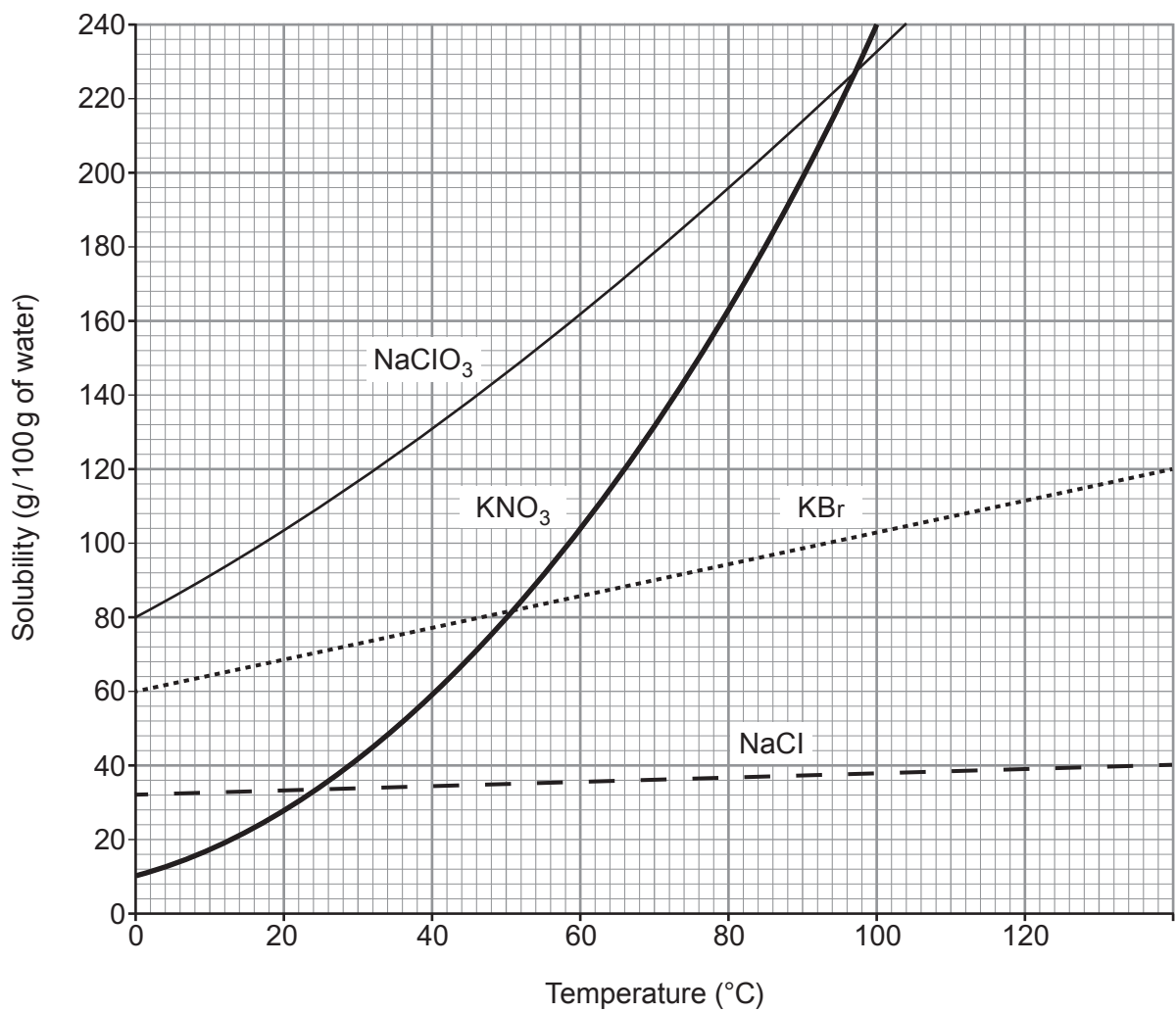


10. The grid below shows the solubility curves for four ionic compounds.



- NaClO<sub>3</sub>    sodium chlorate
- KNO<sub>3</sub>    potassium nitrate
- KBr    potassium bromide
- NaCl    sodium chloride



- (a) (i) Give the temperature at which the solubility of potassium nitrate and potassium bromide is the same. [1]

..... °C

- (ii) Calculate the mass of solid potassium nitrate that would form if a saturated solution in 200 g of water were cooled from 100 °C to 20 °C. [3]

Mass = ..... g

- (iii) Suggest why a student may be surprised at the temperature range shown on the solubility curves. [1]

.....  
.....

- (b) (i) Give the symbols of the **ions** of Group 1 elements present in the compounds shown on the grid. [1]

.....

- (ii) Explain how these ions are formed from their atoms. [2]

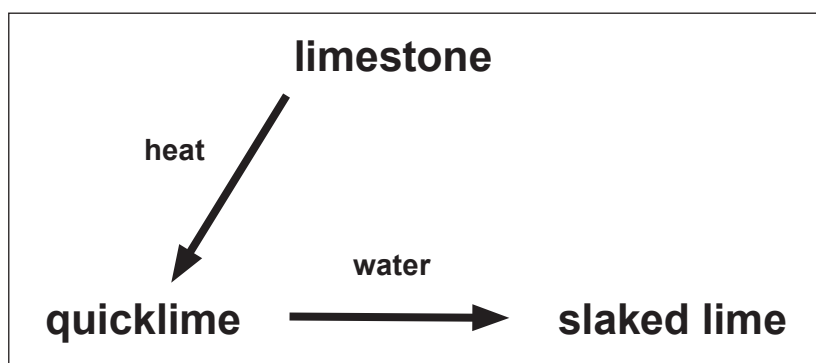
.....  
.....

- (c) Potassium nitrate reacts with aluminium hydroxide to produce aluminium nitrate and potassium hydroxide.

Balance the symbol equation for the reaction taking place. [1]



10. The following diagram shows how limestone,  $\text{CaCO}_3$ , can be converted into useful products.



- (a) When a piece of limestone is heated strongly its mass decreases.

State the type of reaction taking place.

[1]

.....

- (b) (i) Describe what is **seen** when limestone is heated and converted into quicklime. [1]

.....

- (ii) Write a balanced symbol equation for the reaction taking place.

[2]

.....

- (c) (i) Describe what is observed when quicklime is converted into slaked lime. [1]

.....

- (ii) Write a balanced symbol equation for the reaction taking place.

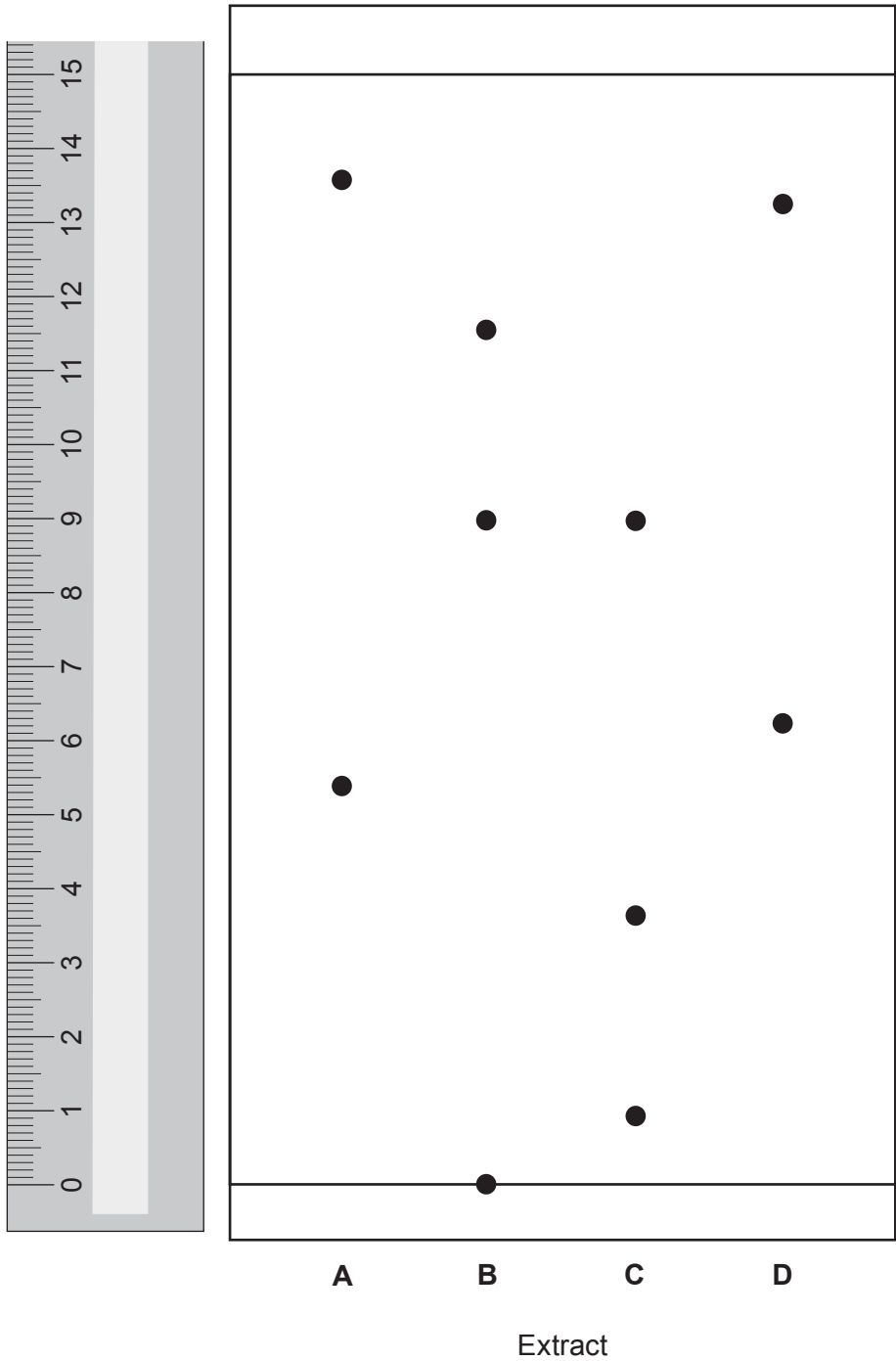
[2]

.....



10. Paper chromatography can be used to identify plant leaf pigments. This process uses a chemical called acetone as the solvent instead of water.

The diagram shows the chromatogram of plant leaf extracts **A**, **B**, **C** and **D** in acetone.



Examiner  
only

(a) All of the extracts contain a mixture of pigments with different  $R_f$  values.

For which plant leaf extract, **A**, **B**, **C** or **D**, is the **highest**  $R_f$  value 0.60?

Give your reasoning. [3]

Extract .....

Reasoning .....

.....  
.....

(b) Explain why the pigments travel different distances on the chromatogram. [2]

.....  
.....  
.....

(c) One of the extracts contains a pigment which is insoluble in acetone.

State the **letter** of this extract. Explain your choice. [2]

Extract .....

Explanation .....

.....

(d) The chemical formula of the solvent acetone is  $C_3H_6O$ . Calculate the percentage by mass of carbon in acetone.

The relative formula mass ( $M_r$ ) of acetone is 58. [2]

$$A_r(C) = 12$$

Percentage = ..... %

|   |
|---|
|   |
| 9 |



10. (a) The table shows information about some Group 1 elements.

| Element   | Relative atomic mass | Number of electrons in the outer shell | Melting point (°C) | Boiling point (°C) | Density (g/cm <sup>3</sup> ) |
|-----------|----------------------|----------------------------------------|--------------------|--------------------|------------------------------|
| lithium   | 7                    | 1                                      | 180                | 1342               | 0.53                         |
| sodium    | 23                   | 1                                      | 98                 | 883                | 0.97                         |
| potassium | 39                   | 1                                      | 63                 | 759                | 0.89                         |
| rubidium  | 85                   | 1                                      | 39                 | 688                | 1.53                         |
| caesium   | 134                  | 1                                      | 29                 | 671                | 1.93                         |

Use the information in the table to answer parts (i) and (ii).

- (i) State the information which explains why the elements have similar chemical properties.

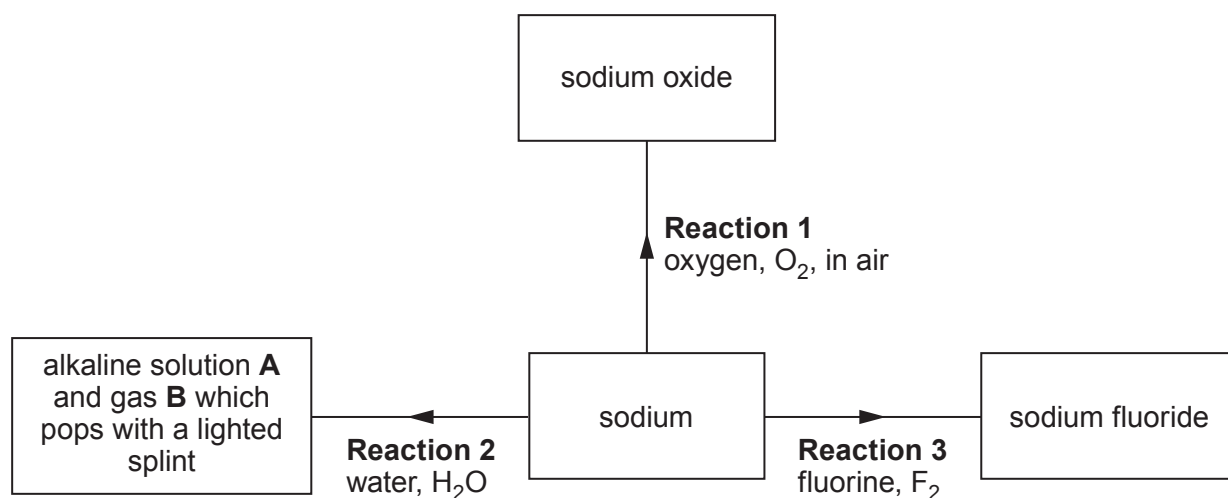
[1]

- (ii) State which **property** has a value which does **not** fit the trend down the group. [1]

.....



(b) The flow diagram shows some reactions of sodium.



(i) State how **Reaction 1** is prevented when storing sodium in the laboratory. [1]

.....

(ii) Give the names of alkaline solution **A** and gas **B**. [2]

..... and .....

(iii) Name the Group 1 metal which would react **least** violently with water. [1]

.....

(iv) Complete the symbol equation for **Reaction 3**. [1]

